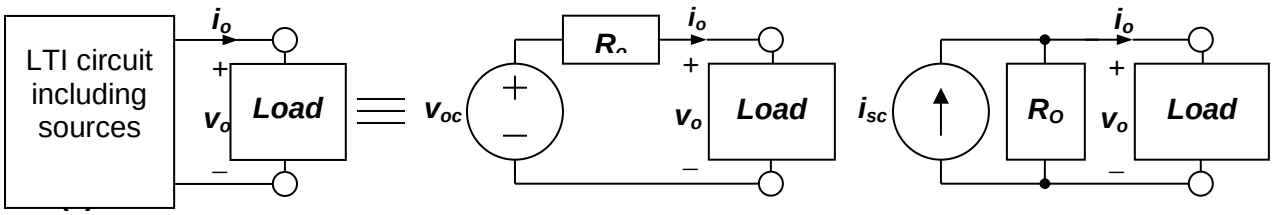
Introduction to Electrical Engineering

Experiment 4 **Thevenin's and Norton's Theorems** **Spring 2019**

Thevenin's and Norton's theorems enable us to construct very simple models to represent a linear time-invariant circuit including independent voltage/current sources, as "seen" by a load connected between any pair of terminals of the circuit. The general structures of Thevenin and Norton equivalents are shown in **Fig. 4.1**. v_{OC} (**Open-Circuit Voltage**) denotes the voltage v_O measured between the terminals when the load is an open-circuit (i_O = 0), and i_{SC} (**Short-Circuit Current**) denotes the current i_O flowing into the load when the load is a short-circuit (v_O = 0). R_o denotes the **Output Resistance** of the circuit, defined to be the resistance measured between the two terminals of the circuit with the load disconnected and all **independent** sources "removed" from within the circuit. It is given by the ratio of the open-circuit voltage v_{OC} and the i_{SC}: $R_o = v_{OC} / i_{SC}$. Thus the Thevenin and Norton equivalents for any given circuit can be obtained if the values of v_{OC} and i_{SC} are measured. But exact open-circuit (i_O = 0) and short-circuit (v_o = 0) conditions are difficult to achieve in practice, and hence one measures v_O and i_O for several values of R_L and determines v_{OC} and i_{SC} by extrapolation.

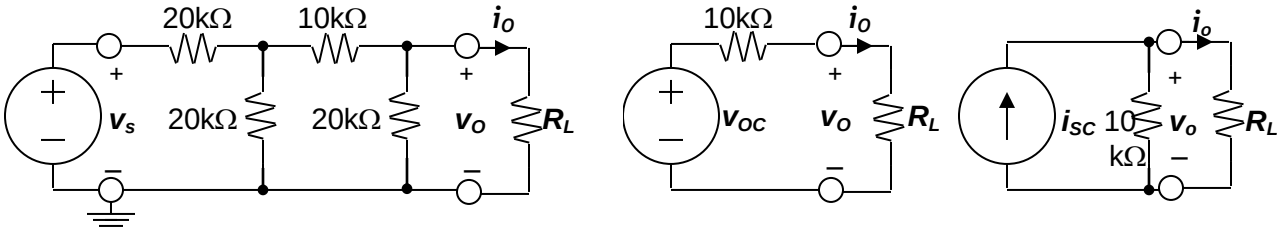


(a) Actual Circuit **(b) Thevenin Equivalent** **(c) Norton Equivalent**

Fig. 4.1 Equivalent circuits based on Thevenin's and Norton's theorems

A. <u>Thevenin and Norton equivalents</u>

1. Assemble the resistive circuit shown in **Fig. 4.2(a)**, where R_L denotes the load resistance and v_s is the WAVEGEN output. Set the WAVEGEN for a **1kHz** sinusoidal output having V_{pp} = 4V and offset = 2V, so that we can expect the voltage v_s to have a minimum level = **0V** and a maximum level = **4V**. However, as WAVEGEN output levels are not very precise, set the Y1 and Y2 cursors precisely at the levels 0V and 4V, and adjust the settings of V_{pp} and offset of the WAVEGEN repetitively to obtain the **minimum** and **maximum** levels of v_s at 0V and 4V **exactly**, as desired.



(a) Actual Circuit **(b) Thevenin Equivalent** **(c) Norton Equivalent**

Fig. 4.2 Resistive circuit with Resistive Load

2. Verify that the circuit of **Fig. 4.2(a)** gets reduced by repeated source transformations to the two equivalents given in **Fig. 4.2(b)** and **Fig. 4.2(c)**, where $v_{OC} = v_s/4$ and $i_{SC} = v_s/(40\text{k}\Omega)$.

3. Measure the **minimum** and **maximum** levels of the output voltage v_O in the circuit of **Fig. 4.2(a)** with the DSO for the following 7 values of the load resistance R_L: 1.00, 2.20, 4.70, 10.0, 22.0, 47.0 and 100 (all in kΩ). Verify that the minimum level is close to zero in all cases, and plot the **peak-to-peak** value of v_O against the **peak-to-peak** value of i_O on a linear graph paper, and draw the best approximation of a straight line passing through the points corresponding to different values of R_L.

4. Determine the **peak-to-peak** values of the **Open-Circuit Voltage** v_{OC} and the **Short-Circuit Current** i_{SC} from the intercepts of the straight line on the two axes, and hence calculate the value of the **Output Resistance** $R_o = v_{OC}/i_{SC}$. Verify that this value is the same as the value indicated in the equivalent circuits of **Fig. 4.2(b)** and **Fig. 4.2(c)**.

B. <u>Thevenin and Norton equivalents with Resistive, Capacitive and Nonlinear Loads</u>

1. Connect a **0.01μF capacitor** as the Load instead of the resistance R_L in the circuit of **Fig. 4.2(a)**. Sketch the input and output waveforms, and measure the **minimum** and **maximum** levels of the output voltage v_O.

2. Connect the given **diode** as the Load instead of the resistance R_L in the circuit of **Fig. 4.2(a)**. Choose the polarity of the diode so as to ensure the flow of current through the diode. Sketch the input and output waveforms, and measure the **minimum** and **maximum** levels of the output voltage v_O

3. Now set up the Thevenin equivalent circuit given in **Fig. 4.2(b)**, with a Load Resistance R_L = 10kΩ, and adjust the **maximum** level of the WAVEGEN output to **1V**, keeping the **minimum** level at **0V**, so that the input voltage is the same as expected value of v_{OC} (= v_s / 4). Measure the **minimum** and **maximum** levels of the output voltage v_O, and compare these values with the values measured in step **A.3** for the same Load Resistance (R_L = 10kΩ)

4. Now connect the same **0.01μF capacitor** as used in step **B.1** as the Load instead of the 10kΩ resistor in the circuit of **Fig. 4.2(b)**. Sketch the input and output waveforms, and measure the **minimum** and **maximum** levels of the output voltage v_O. Compare these measured values with those obtained in step **B.1**.

5. Now connect the same **diode** as used in step **B.2** as the Load instead of the 10kΩ resistor in the circuit of **Fig. 4.2(b)**. Sketch the input and output waveforms, and measure the **minimum** and **maximum** levels of the output voltage v_O. Compare these measured values with those obtained in step **B.2**.